UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

# Lagrange's Method of Undetermined Multipliers

Complete study pack for class 011: extrema subject to a constraint, the auxiliary function $F = f + \lambda\phi$, the working procedure, its advantages and limits, and every geometric and applied problem asked in the ESA

Compiled from the PESU class 011 slide deck, its two AV-summary transcripts, the Unit 1 class-work sheet, an extra question list and past ESA papers. Generated 15 October 2026.

# · How to use this pack

Topic 1 is the method: learn the procedure (Figure F1.1) and how to pick $f$ and $\phi$ (Figure F1.2). Topic 2 works the geometric problems (inscribed solids, nearest and farthest points), topic 3 the applied ones (boxes, splitting a number, temperature). Lagrange's method was asked in five of the eight past papers, so topics 2–3 carry the most marks in Unit 1 after Taylor's series. Free maxima and minima (the $rt - s^2$ test) are classes 009–010. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S011·5 | Deck 011, slide 5 (rendered .pptx) |
| T011a 00:01:08 | Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start |
| T011b 00:05:03 | Transcript “Problems on Lagrange's method of undetermined multipliers” |
| CW U1 P33 | Unit 1 class-work problem 33 |
| Extra Q57 | Extra question list “lagrange-ques”, question 57 |
| ESA Jul-23 1c | Past paper: Jul 2023, question 1.c |

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4. Appendix A · Symbol glossary
5. Appendix B · Sources, coverage and gaps

# 1 Lagrange's method: the auxiliary function and the procedure

To find the extreme values of $f(x, y, z)$ subject to a constraint $\phi(x, y, z) = 0$, introduce an unknown constant $\lambda$ (the Lagrange multiplier) and form $F = f + \lambda\phi$. Solve the four equations $F_x = 0$, $F_y = 0$, $F_z = 0$, $\phi = 0$ for $x, y, z, \lambda$. No variable has to be eliminated first, and the method extends to more variables and constraints; but it gives only stationary points, not their nature.

### Key terms and definitions

**Constraint.** A condition $\phi(x, y, z) = 0$ that the variables must satisfy (walk only on the surface of a sphere, a fixed surface area, a fixed sum). S011·3 · T011a 00:00:36

**Lagrange multiplier.** Suppose we need to minimize or maximize a function $f(x, y, z)$ subject to the constraint $\phi(x, y, z)$; then we can introduce an additional unknown constant $\lambda$, known as Lagrange's multiplier, to ease the process of finding the extrema. S011·3 · T011a 00:00:53

**Auxiliary function.** $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$: the given function plus $\lambda$ times its constraint (also called Lagrange's function). S011·3 · T011a 00:01:08 · T011b 00:02:06

> *"Working procedure. Step 1: Form the auxiliary equation $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$ … (1). Step 2: Partially differentiate $F$ in (1) w.r.t. $x, y, z$ respectively. Step 3: Solve the four equations $F_x = 0$, $F_y = 0$, $F_z = 0$ and the constraint for the Lagrange multiplier $\lambda$ and stationary values $x, y, z$."* S011·3

> *"Advantages: 1. The stationary points of $f(x, y, z)$ can be determined without determining $x, y, z$ explicitly. 2. This method can be extended to function of several variables subject to many constraints. Disadvantages: 1. Nature of the stationary points can not be determined. 2. Equations $F_x = 0$, $F_y = 0$, $F_z = 0$ are only necessary conditions."* S011·4 · T011a 00:03:04

### Formulas

**E1.1 · Auxiliary (Lagrange) function** ESA ★★★

$$F(x, y, z) = f(x, y, z) + \lambda\,\phi(x, y, z)$$

| | |
|---|---|
| gives | One function whose stationary points are the constrained stationary points. |
| applies | Optimise $f$ subject to $\phi(x, y, z) = 0$. |
| note | Write the constraint with everything on one side: $x^2 + y^2 + z^2 - r^2$, $xy + 2yz + 2zx - 432$. |

S011·3 · T011a 00:01:08

### E1.2 · The four equations

ESA ★★★

$$F_x = f_x + \lambda\phi_x = 0, \quad F_y = f_y + \lambda\phi_y = 0, \quad F_z = f_z + \lambda\phi_z = 0, \quad \phi = 0$$

GIVES $x, y, z$ and $\lambda$ (four unknowns, four equations).

APPLIES Always; with two variables drop $F_z$ (three equations).

NOTE Equivalently $\nabla f = -\lambda\nabla\phi$: the gradients are parallel (added for clarity).

S011·3 · T011a 00:02:17 · T011a 00:02:30

### E1.3 · Solving trick: make $\lambda$ the subject

ESA ★★

$$\lambda = -\frac{f_x}{\phi_x} = -\frac{f_y}{\phi_y} = -\frac{f_z}{\phi_z}$$

GIVES Relations between $x, y, z$ without $\lambda$.

APPLIES When $\phi_x, \phi_y, \phi_z \neq 0$.

NOTE Or multiply the equations by $x$, $y$, $z$ and subtract (slides 011·6, 011·11).

S011·13 · S011·14 · T011b 00:07:24

## Graphs and diagrams

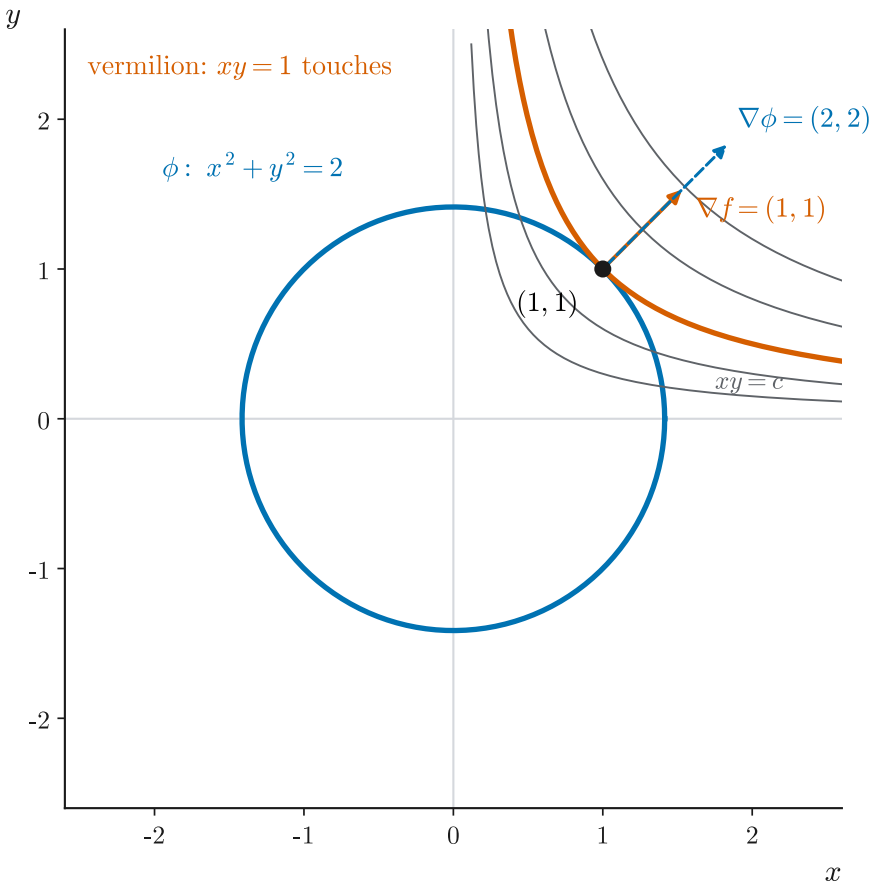
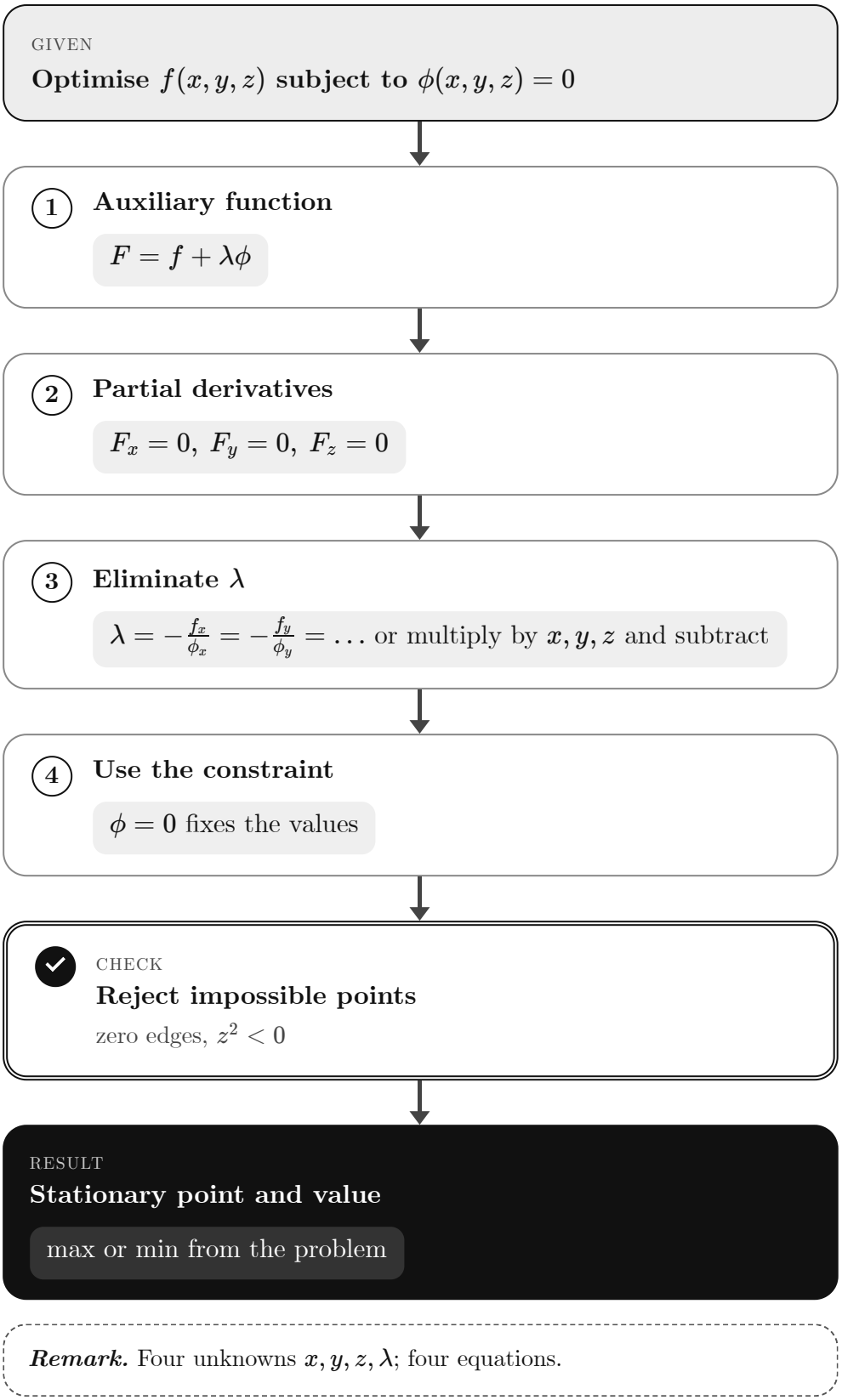



**Figure G1.1.** Why the method works: a level curve touches the constraint. ESA ★

ADDED FOR CLARITY S011·3 · S011·7

***How to read it.*** Maximise $f = xy$ on the circle $x^2 + y^2 = 2$ (blue). The level curves $xy = c$ (grey hyperbolas) cross the circle until $c = 1$, where $xy = 1$ (vermilion) just touches it at $(1, 1)$. There $\nabla f$ and $\nabla\phi$ point the same way (arrows): $\nabla f + \lambda\nabla\phi = 0$ with $\lambda = -\frac{1}{2}$.

## Flowcharts




***Remark.*** Four unknowns $x, y, z, \lambda$; four equations.

**Figure F1.1.** Lagrange's working procedure ESA ★★★

S011·3 · T011a 00:01:18

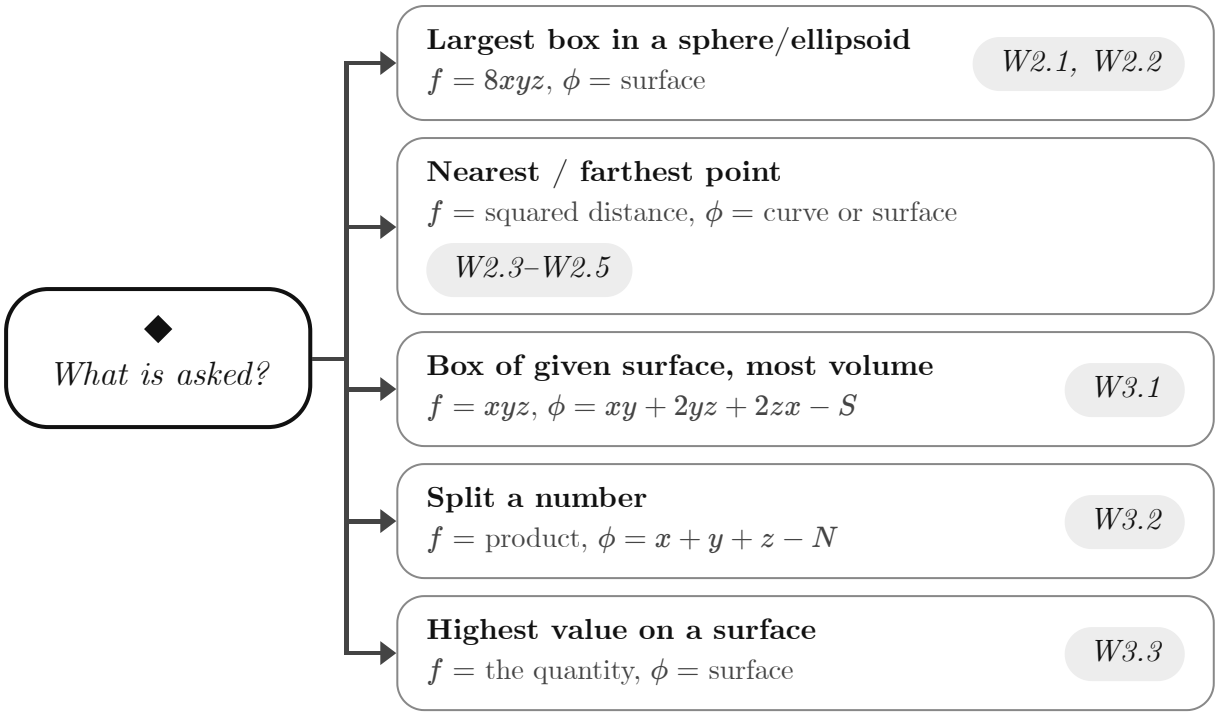



**Figure F1.2.** Choosing $f$ and $\phi$ ESA ★★★

T011a 00:09:15 · T011b 00:01:46 · S011·10 · S011·12

## Elimination (class 010) and Lagrange's method

| | Elimination | Lagrange's multipliers |
|---|---|---|
| **Constraint used** | Solved for one variable and substituted | Added as $\lambda\phi$ |
| **Equations** | $f_x = f_y = 0$ in two variables | $F_x = F_y = F_z = 0$, $\phi = 0$ |
| **Nature of the point** | $rt - s^2$ and $r$ decide | Not decided; from the problem |
| **More variables / constraints** | Hard | Extends directly |
| **Temperature $400xyz^2$** | 50 (class 010) | 50 (W3.3) |

S011·4 · T011a 00:01:36 · T011a 00:04:23

## Important points and why

1. The method avoids elimination: in class 010 we had to substitute $z^2 = 1 - x^2 - y^2$ for the temperature problem; with Lagrange we differentiate $F$ directly. ESA ★★ T011a 00:01:18 · T011a 00:03:17
2. Four unknowns $x, y, z, \lambda$ need four equations: $F_x = 0$, $F_y = 0$, $F_z = 0$ and the constraint $\phi = 0$. ESA ★★★ T011a 00:02:30
3. $\lambda$ is "undetermined": we may find it, but it does not appear in the answer. ESA ★ T011a 00:02:04

4. The method gives only stationary points; to verify maximum or minimum by $rt - s^2$ we would have to go back to the previous method. In problems the question itself says which (maximum volume, closest point). ESA ★★ S011·4 · T011a 00:04:05 · T011a 00:14:01
5. The most important step is to identify the function $f$ and the constraint $\phi$ correctly (Figure F1.2). ESA ★★★ T011a 00:09:15 · T011b 00:01:46
6. Geometrically, at a constrained extremum a level curve (surface) of $f$ just touches the constraint, so $\nabla f$ and $\nabla \phi$ are parallel: $\nabla f = -\lambda \nabla \phi$ (Figure G1.1). ESA ★ ADDED FOR CLARITY S011·7

**Why does $F_x = F_y = F_z = 0$ find the constrained extrema?**
At a constrained extremum, $f$ cannot increase along any direction that stays on the surface $\phi = 0$, so $\nabla f$ is perpendicular to the surface, i.e. parallel to $\nabla \phi$: $\nabla f + \lambda \nabla \phi = 0$, which is exactly $F_x = F_y = F_z = 0$.
ADDED FOR CLARITY S011·3 · S011·7

***Exam trap.*** Do not forget the constraint as the fourth equation; three equations cannot fix four unknowns. T011a 00:02:30

***Exam trap.*** Dividing by $x$, $y$, $z$ or $\lambda$ assumes it is not zero; check those cases separately (they usually give zero volume or an impossible point). ADDED FOR CLARITY S011·9 · T011b 00:03:40

SHORT-NOTE RECAP

Optimise $f$ subject to $\phi = 0$: $F = f + \lambda \phi$; solve $F_x = F_y = F_z = 0$ with $\phi = 0$ for $x, y, z, \lambda$. Advantages: no elimination; extends to many variables and constraints. Disadvantages: nature not decided; the equations are only necessary.

# 2 Geometric problems: inscribed solids, nearest and farthest points

The largest rectangular solid in a sphere is a cube; in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ its half-edges are $\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}$ and its volume $\frac{8abc}{3\sqrt{3}}$. For nearest and farthest points, optimise the square of the distance: the hyperbolic cylinder $x^2-z^2=1$ is nearest the origin at $(\pm 1,0,0)$; the curve $5x^2+6xy+5y^2=8$ is between distances 1 and 2.

### Key terms and definitions

**Continued product.** The product obtained by multiplying together in succession: for parts $x, y, z$, the product of the first, the square of the second and the cube of the third is $xy^2z^3$. S011·12 · T011b 00:05:23

### Formulas

**E2.1 · Largest box in a sphere and in an ellipsoid** ESA ★★★

$$x^2+y^2+z^2=r^2:\ x=y=z=\tfrac{r}{\sqrt{3}},\ V_{\max}=\tfrac{8r^3}{3\sqrt{3}}$$
$$\tfrac{x^2}{a^2}+\tfrac{y^2}{b^2}+\tfrac{z^2}{c^2}=1:\ x=\tfrac{a}{\sqrt{3}},\ y=\tfrac{b}{\sqrt{3}},\ z=\tfrac{c}{\sqrt{3}},\ V_{\max}=\tfrac{8abc}{3\sqrt{3}}$$

| | |
|---|---|
| gives | Half-edges and the largest volume $V=8xyz$. |
| applies | Box centred at the origin, edges $2x, 2y, 2z$ parallel to the axes. |
| note | The sphere is the ellipsoid with $a=b=c=r$: the box is a cube. |

S011·5 · ESA May-24 1c · ESA Jan-24 s3 1a · ESA Jan-24 s4 1a

**E2.2 · Distance problems: use the square** ESA ★★★

$$f=x^2+y^2+z^2\ \text{(origin)},\qquad f=(x-x_0)^2+(y-y_0)^2+(z-z_0)^2$$

| | |
|---|---|
| gives | The squared distance, which has its extrema at the same points as the distance. |
| applies | Nearest or farthest point on a curve or surface. |
| note | Take the square root at the end. |

added for clarity S011·8 · ESA Jul-23 1c · CW U1 P36

## Graphs and diagrams

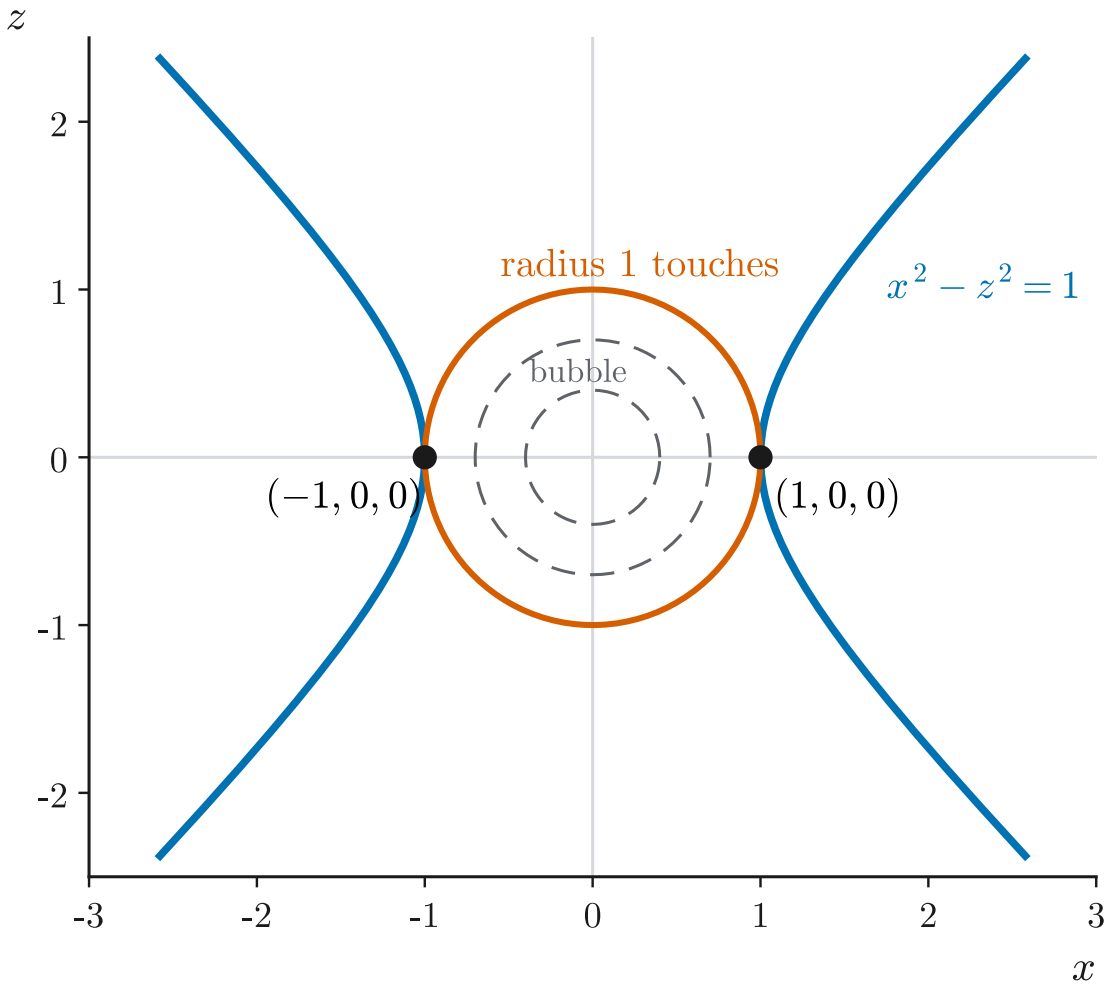



**Figure G2.1.** The soap bubble: $x^2 - z^2 = 1$ nearest the origin. ESA ★★

S011·7 · S011·9

***How to read it.*** Cross-section $y = 0$. The hyperbolic cylinder appears as the hyperbola $x^2 - z^2 = 1$ (blue). Circles centred at the origin (grey, growing like a soap bubble) first touch it at $(\pm 1, 0)$ with radius 1 (vermilion): the nearest points $(\pm 1, 0, 0)$.

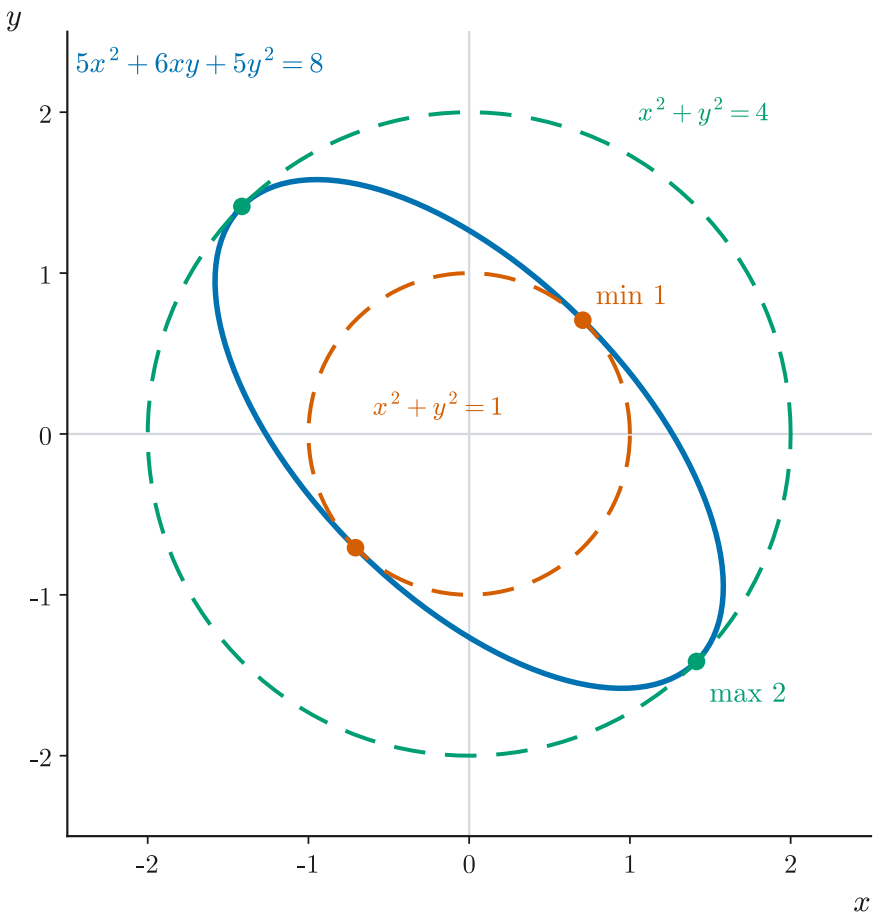



**Figure G2.2.** $5x^2 + 6xy + 5y^2 = 8$ between the circles of radius 1 and 2. ESA ★★★

ADDED FOR CLARITY ESA Jul-23 1c

***How to read it.*** The curve is an ellipse (blue) tilted at $45°$. The circle $x^2 + y^2 = 1$ (vermilion) touches it from inside at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$: minimum distance 1. The circle $x^2 + y^2 = 4$ (green) touches it from outside at $\pm(\sqrt{2}, -\sqrt{2})$: maximum distance 2.

## Important points and why

1. For a box inscribed in a sphere or ellipsoid centred at the origin, take the corner $(x, y, z)$ in the first octant: edges $2x, 2y, 2z$, volume $8xyz$. Using $xyz$ instead gives the same point. ESA ★★★ S011·5 · T011a 00:06:08 · T011a 00:06:53
2. To find the nearest point, imagine a small sphere centred at the origin expanding like a soap bubble until it just touches the surface (Figure G2.1). ESA ★ S011·7
3. Reject solutions that are impossible: on $x^2 - z^2 = 1$, $x = 0$ would need $z^2 = -1$. ESA ★★ S011·9 · T011a 00:12:26
4. When there are two stationary values (W2.4: distances 1 and 2), the larger is the maximum and the smaller the minimum on a closed curve. ESA ★★★ ADDED FOR CLARITY ESA Jul-23 1c

**Why minimise $x^2+y^2+z^2$ instead of the distance?**
$\sqrt{u}$ increases with $u$, so both have their extrema at the same points, and the square has no square root to differentiate.

ADDED FOR CLARITY S011·8 · T011a 00:10:08

***Exam trap.*** The box in the sphere has edges $2x, 2y, 2z$, so $V = 8xyz$: its largest volume is $\frac{8r^3}{3\sqrt{3}}$, not $\frac{r^3}{3\sqrt{3}}$. ADDED FOR CLARITY S011·5

***Exam trap.*** Slide 011·6's third line should read $8xyz + 2z^2\lambda = 0$ (printed with $y^2$).

ADDED FOR CLARITY S011·6

## Worked examples

### W2.1 · The largest rectangular solid in a sphere is a cube — ESA ★★★

**GIVEN** Show that the rectangular solid of maximum volume that can be inscribed in a given sphere $x^2+y^2+z^2=r^2$ is a cube.

**STEP 1** Edges $2x, 2y, 2z$: $f = 8xyz$; $\phi = x^2+y^2+z^2-r^2$; $F = 8xyz + \lambda(x^2+y^2+z^2-r^2)$.

**STEP 2** $F_x = 8yz + 2x\lambda = 0$, $F_y = 8xz + 2y\lambda = 0$, $F_z = 8xy + 2z\lambda = 0$.

**STEP 3** Multiply by $x$, $y$, $z$: $8xyz + 2x^2\lambda = 0$, $8xyz + 2y^2\lambda = 0$, $8xyz + 2z^2\lambda = 0$.

**STEP 4** So $2x^2\lambda = 2y^2\lambda = 2z^2\lambda = -8xyz \neq 0$: $x^2 = y^2 = z^2$, and with $x, y, z > 0$, $x = y = z$.

**STEP 5** From the constraint $3x^2 = r^2$: $x = y = z = \frac{r}{\sqrt{3}}$.

**ANSWER** $\boxed{x = y = z\text{: the solid is a cube of edge } \dfrac{2r}{\sqrt{3}}\text{, volume } \dfrac{8r^3}{3\sqrt{3}}}$

**CHECK** sympy solves the system to $x = y = z = \frac{r}{\sqrt{3}}$ and $V = \frac{8\sqrt{3}r^3}{9} = \frac{8r^3}{3\sqrt{3}}$.

S011·5 · S011·6 · S011·14 · T011a 00:04:49 · T011b 00:08:50 · ESA May-24 1c · Extra Q59

## W2.2 · Largest parallelepiped in an ellipsoid ESA ★★★

**GIVEN** Use the method of Lagrange's multipliers to find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

**STEP 1** $f = 8xyz$, $F = 8xyz + \lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1\right)$.

**STEP 2** $F_x = 8yz + \frac{2\lambda x}{a^2} = 0$, $F_y = 8xz + \frac{2\lambda y}{b^2} = 0$, $F_z = 8xy + \frac{2\lambda z}{c^2} = 0$.

**STEP 3** Multiply by $x$, $y$, $z$: $\frac{2\lambda x^2}{a^2} = \frac{2\lambda y^2}{b^2} = \frac{2\lambda z^2}{c^2} = -8xyz$, so $\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2}$.

**STEP 4** Constraint: $3\frac{x^2}{a^2} = 1$, so $x = \frac{a}{\sqrt{3}}$, $y = \frac{b}{\sqrt{3}}$, $z = \frac{c}{\sqrt{3}}$.

**STEP 5** $V = 8 \cdot \frac{abc}{3\sqrt{3}}$.

**ANSWER** $\boxed{V_{\max} = \dfrac{8abc}{3\sqrt{3}} \text{ (edges } \tfrac{2a}{\sqrt{3}}, \tfrac{2b}{\sqrt{3}}, \tfrac{2c}{\sqrt{3}})}$

**CHECK** sympy: $\frac{8\sqrt{3}abc}{9}$, the same; with $a = b = c = r$ it reduces to W2.1.

ESA Jan-24 s3 1a · ESA Jan-24 s4 1a · Extra Q61

## W2.3 · Points of $x^2 - z^2 = 1$ closest to the origin ESA ★★

**GIVEN** Find the points closest to the origin on the hyperbolic cylinder $x^2 - z^2 = 1$.

**STEP 1** $f = x^2 + y^2 + z^2$ (squared distance), $\phi = x^2 - z^2 - 1$; $F = x^2 + y^2 + z^2 + \lambda(x^2 - z^2 - 1)$.

**STEP 2** $F_x = 2x(1+\lambda) = 0$ (1), $F_y = 2y = 0$ (2), $F_z = 2z(1-\lambda) = 0$ (3), $x^2 - z^2 = 1$ (4).

**STEP 3** (2): $y = 0$. (1): $x = 0$ or $\lambda = -1$; $x = 0$ would need $z^2 = -1$, so $\lambda = -1$.

**STEP 4** (3) with $\lambda = -1$: $4z = 0$, $z = 0$. (4): $x^2 = 1$, $x = \pm 1$.

**ANSWER** $\boxed{(\pm 1, 0, 0)\text{, at distance 1 from the origin}}$

**CHECK** sympy finds only these two points ($\lambda = -1$).

S011·7 · S011·8 · S011·9 · T011a 00:09:35

### W2.4 · Maximum and minimum distances from the origin to $5x^2 + 6xy + 5y^2 = 8$ ESA ★★★

GIVEN Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.

STEP 1 $f = x^2 + y^2$, $F = x^2 + y^2 + \lambda(5x^2 + 6xy + 5y^2 - 8)$.

STEP 2 $F_x = 2x + \lambda(10x + 6y) = 0$, $F_y = 2y + \lambda(6x + 10y) = 0$.

STEP 3 Eliminate $\lambda$: $\frac{x}{5x+3y} = \frac{y}{3x+5y}$, so $3x^2 + 5xy = 5xy + 3y^2$, $x^2 = y^2$, $y = \pm x$.

STEP 4 $y = x$: $16x^2 = 8$, $x^2 = \frac{1}{2}$; $f = x^2 + y^2 = 1$: distance 1, at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

STEP 5 $y = -x$: $4x^2 = 8$, $x^2 = 2$; $f = 4$: distance 2, at $\pm(\sqrt{2}, -\sqrt{2})$.

ANSWER Maximum distance $2$, minimum distance $1$

CHECK sympy agrees. In rotated axes the curve is the ellipse $8u^2 + 2v^2 = 8$ with semi-axes 1 and 2 (Figure G2.2).

ESA Jul-23 1c

### W2.5 · Distances of $(3, 4, 12)$ from the unit sphere ESA ★★

GIVEN Find the maximum and minimum distances of the point $(3, 4, 12)$ from the unit sphere $x^2 + y^2 + z^2 = 1$.

STEP 1 $f = (x-3)^2 + (y-4)^2 + (z-12)^2$, $F = f + \lambda(x^2 + y^2 + z^2 - 1)$.

STEP 2 $2(x-3) + 2\lambda x = 0 \Rightarrow x = \frac{3}{1+\lambda}$; likewise $y = \frac{4}{1+\lambda}$, $z = \frac{12}{1+\lambda}$.

STEP 3 Constraint: $\frac{9+16+144}{(1+\lambda)^2} = 1$, $1 + \lambda = \pm 13$: points $\pm\frac{1}{13}(3, 4, 12)$.

STEP 4 Distances: $13 - 1 = 12$ and $13 + 1 = 14$.

ANSWER Maximum $14$, minimum $12$

CHECK CW answer agrees; sympy agrees.

CW U1 P36 · Extra Q45

## Past-paper questions

ESA Jul 2023 (7 marks) UE22MA141A Q1.c · solved in W2.4

Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.

ESA May 2024 (7 marks) UE22MA141A Q1.c · solved in W2.1

Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube by using Lagrange's method of multipliers.

ESA Jan 2024 7 marks UE23MA141A set 3, Q1.a · solved in W2.2

Use the method of Lagrange's multipliers to find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

ESA Jan 2024 7 marks UE23MA141A set 4, Q1.a · solved in W2.2

The same question as set 3 Q1.a (largest parallelepiped in the ellipsoid).

SHORT-NOTE RECAP

Box in sphere: $f=8xyz \Rightarrow x=y=z=\frac{r}{\sqrt{3}}$, cube. Ellipsoid: $V_{\max}=\frac{8abc}{3\sqrt{3}}$. Distances: optimise the square. $x^2-z^2=1$: nearest $(\pm1,0,0)$. $5x^2+6xy+5y^2=8$: distances 1 (min) and 2 (max). $(3,4,12)$ to unit sphere: 12 and 14.

# 3 Applied problems: boxes, splitting a number, temperature

The same procedure handles every applied problem once $f$ and $\phi$ are identified: an open box of surface 432 cm² holds most as $12 \times 12 \times 6$; 24 split as $4 + 8 + 12$ maximises $xy^2z^3$; the temperature $400xyz^2$ on the unit sphere is at most 50. The trick in solving is to make $\lambda$ the subject of each equation and equate.

### Important points and why

1. Open box (no top): surface $xy + 2yz + 2zx$; closed box: $2(xy + yz + zx)$. ESA ★★ S011·10 · T011b 00:01:09
2. Dividing a number $N$ into parts $x, y, z$: constraint $x + y + z = N$; the function is the product asked for. ESA ★ S011·12 · T011b 00:05:57
3. The temperature problem of class 010 gives the same 50 by Lagrange's method with no elimination (W3.3). ESA ★★★ T011a 00:14:33 · ESA Jun-24 1d

***Exam trap.*** Open-top box: $x = y = 2z$ (height half the base edge), not a cube. S011·11 · CW U1 P34

### Worked examples

**W3.1 · Open box of surface 432 cm² with maximum capacity** ESA ★★

**given** Find the dimensions of the rectangular box, open at the top, of maximum capacity whose surface is 432 sq cm.

**step 1** $f = xyz$, $\phi = xy + 2yz + 2zx - 432$; $F = xyz + \lambda(xy + 2yz + 2zx - 432)$.

**step 2** $yz + \lambda(y + 2z) = 0$ (1), $xz + \lambda(x + 2z) = 0$ (2), $xy + \lambda(2y + 2x) = 0$ (3).

**step 3** $x\cdot(1) - y\cdot(2)$: $2z\lambda(x - y) = 0 \Rightarrow x = y$. $y\cdot(2) - z\cdot(3)$: $x\lambda(y - 2z) = 0 \Rightarrow y = 2z$.

**step 4** Constraint with $x = y = 2z$: $x^2 + x^2 + x^2 = 3x^2 = 432$, $x = 12$.

**answer** $12$ cm $\times$ $12$ cm $\times$ $6$ cm (capacity $864$ cm³)

**check** sympy agrees.

S011·10 · S011·11 · T011b 00:00:19

### W3.2 · Divide 24 to maximise $xy^2z^3$ ESA ★★

**GIVEN** Divide the number 24 into three parts such that the continued product of the first, square of the second and cube of the third may be maximum.

**STEP 1** $f = xy^2z^3$, $\phi = x + y + z - 24$; $F = xy^2z^3 + \lambda(x + y + z - 24)$.

**STEP 2** $\lambda = -y^2z^3 = -2xyz^3 = -3xy^2z^2$.

**STEP 3** $y^2z^3 = 2xyz^3 \Rightarrow y = 2x$; $2xyz^3 = 3xy^2z^2 \Rightarrow z = \frac{3}{2}y = 3x$.

**STEP 4** $x + 2x + 3x = 24$: $x = 4$.

**ANSWER** Parts $4$, $8$, $12$ (product $4 \cdot 8^2 \cdot 12^3 = 442368$)

**CHECK** sympy agrees.

S011·12 · S011·13 · T011b 00:05:03 · Extra Q53

### W3.3 · Highest temperature $400xyz^2$ on the unit sphere ESA ★★★

**GIVEN** The temperature $T$ at any point $(x, y, z)$ in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

**STEP 1** $F = 400xyz^2 + \lambda(x^2 + y^2 + z^2 - 1)$.

**STEP 2** $400yz^2 + 2\lambda x = 0$, $400xz^2 + 2\lambda y = 0$, $800xyz + 2\lambda z = 0$.

**STEP 3** Multiply by $x$, $y$, $z$: $400xyz^2 = -2\lambda x^2 = -2\lambda y^2$ and $800xyz^2 = -2\lambda z^2$.

**STEP 4** So $x^2 = y^2$ and $z^2 = 2x^2$. Constraint: $x^2 + x^2 + 2x^2 = 1$, $x^2 = \frac{1}{4}$, $z^2 = \frac{1}{2}$.

**STEP 5** With $x, y$ of the same sign: $T = 400 \cdot \frac{1}{4} \cdot \frac{1}{2} = 50$ (opposite signs give $-50$).

**ANSWER** Highest temperature $50$, at $x = y = \pm\frac{1}{2}$, $z = \pm\frac{1}{\sqrt{2}}$

**CHECK** sympy: stationary values $\{0, 50, -50\}$. Same as class 010's elimination answer.

ESA Jun-24 1d · Extra Q57 · T011a 00:00:17

### W3.4 · Extreme value of $a^3x^2 + b^3y^2 + c^3z^2$ ESA ★★

**GIVEN** Find the extreme value of $a^3x^2 + b^3y^2 + c^3z^2$ subject to $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$, where $a, b, c > 0$.

**STEP 1** $F = a^3x^2 + b^3y^2 + c^3z^2 + \lambda\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 1\right)$.

**STEP 2** $2a^3x - \frac{\lambda}{x^2} = 0 \Rightarrow a^3x^3 = \frac{\lambda}{2}$; likewise $b^3y^3 = c^3z^3 = \frac{\lambda}{2}$. So $ax = by = cz = t$.

**STEP 3** Constraint: $\frac{a}{t} + \frac{b}{t} + \frac{c}{t} = 1$, $t = a + b + c$.

**STEP 4** Value: $a^3\frac{t^2}{a^2} + b^3\frac{t^2}{b^2} + c^3\frac{t^2}{c^2} = t^2(a + b + c) = t^3$.

**ANSWER** $\boxed{(a + b + c)^3 \text{ at } \left(\frac{t}{a}, \frac{t}{b}, \frac{t}{c}\right),\ t = a + b + c}$

**CHECK** CW answer agrees; numerically with $a, b, c = 1, 2, 3$: value $216 = 6^3$.

CW U1 P33

### W3.5 · Extreme value of $x^3 + 8y^3 + 64z^3$ when $xyz = 1$ ESA ★

**GIVEN** Find the extreme value of $x^3 + 8y^3 + 64z^3$ subject to $xyz = 1$.

**STEP 1** $F = x^3 + 8y^3 + 64z^3 + \lambda(xyz - 1)$: $3x^2 + \lambda yz = 0$, $24y^2 + \lambda xz = 0$, $192z^2 + \lambda xy = 0$.

**STEP 2** Multiply by $x$, $y$, $z$: $3x^3 = 24y^3 = 192z^3 = -\lambda$, so $x = 2y = 4z$.

**STEP 3** $xyz = x \cdot \frac{x}{2} \cdot \frac{x}{4} = 1$: $x^3 = 8$, $x = 2$, $y = 1$, $z = \frac{1}{2}$.

**ANSWER** $\boxed{24 \text{ at } \left(2, 1, \frac{1}{2}\right) \text{ (a minimum for positive } x, y, z)}$

**CHECK** CW answer agrees; sympy agrees.

CW U1 P35

## Past-paper questions

ESA June 2024 · 7 marks · UE23MA141A Q1.d · solved in W3.3

The temperature $T$ at any point $(x, y, z)$ in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

**SHORT-NOTE RECAP**

Open box, surface 432: $12 \times 12 \times 6$. $24 = 4 + 8 + 12$ for $xy^2z^3$. $400xyz^2$ on the unit sphere: 50. CW: $a^3x^2 + b^3y^2 + c^3z^2$ with $\sum \frac{1}{x} = 1$: $(a + b + c)^3$; $x^3 + 8y^3 + 64z^3$ with $xyz = 1$: 24.

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|
| $f(x, y, z)$ | Objective | Function to maximise or minimise | E1.1 |
| $\phi(x, y, z)$ | Constraint | $\phi = 0$; printed $\varnothing$ on the slides | E1.1, E1.2 |
| $\lambda$ | Lagrange multiplier | Undetermined constant; not in the answer | E1.1–E1.3 |
| $F$ | Auxiliary function | $f + \lambda\phi$ | E1.1 |
| $r$ | Radius | Of the sphere $x^2 + y^2 + z^2 = r^2$; not $f_{xx}$ | E2.1 |
| $a,\ b,\ c$ | Semi-axes | Of the ellipsoid (in W3.4: positive constants) | E2.1, W3.4 |
| $2x,\ 2y,\ 2z$ | Edges | Of the inscribed box; volume $8xyz$ | E2.1 |

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Deck 011 (.pptx rendered; .slides.md; .slides.txt) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.* | 15 slides; content on 3–14 |
| Transcript T011a (all 4 formats) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.* | 14.8 min: method, cube in a sphere, hyperbolic cylinder |
| Transcript T011b (all 4 formats) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.* | 10.2 min: open box, dividing 24, cube in a sphere |
| Unit 1 & 2 class-work sheet (UE24MA141A) | pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf | Problems 33–36 (with answers) |
| Extra questions “lagrange-ques” (third-party list) | pesuc/First_Year_8enterprise/Files/Math 1/Extra/Extra Questions/lagrange-ques__1smrsCddi3.pdf | Confirms ESA problems (Q57, Q59, Q61); no solutions used |
| ESA papers, UE22MA141A and UE23MA141A (8 papers) | pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf | 5 in-scope Q1 parts (4 distinct problems) |
| Videos 011 (.mp4) | pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_*.mp4 | Not opened: every slide is readable and every result was recomputed |

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Lagrange's method is asked in five of them, always as a 7-mark Q1 part, with four distinct problems: Jul 2023 Q1.c (maximum and minimum distances from the origin to $5x^2 + 6xy + 5y^2 - 8 = 0$), May 2024 Q1.c (the rectangular solid of maximum volume in a sphere is a cube; also slides 011·5–6), Jan 2024 sets 3 and 4 Q1.a (largest parallelepiped in an ellipsoid) and June 2024 Q1.d (highest temperature $400xyz^2$ on the unit sphere; also class 010 by elimination). Three of the four also appear in the extra question list (Q57, Q59, Q61). Dec 2023 and Jan 2024 sets 1–2 ask free extrema (class 010) instead.

### Known gaps, conflicts and corrections

1. The deck is titled “... for two variables”, but the method, the slides' problems and all four ESA problems use three variables $x, y, z$ and one constraint; the two-variable case ($F_x = F_y = 0$, $\phi = 0$) is the same with $z$ dropped (Jul 2023 Q1.c, W2.4).
2. Slide 011·6 prints the third line as $8xzy + 2y^2\lambda = 0$; multiplying $8xy + 2z\lambda = 0$ by $z$ gives $8xyz + 2z^2\lambda = 0$. The conclusion $2x^2\lambda = 2y^2\lambda = 2z^2\lambda$, so $x = y = z$, is right.
3. Slide 011·9 says “from (3), we get $z = 0$, $\lambda = 1$”; equation (3), $2z(1 - \lambda) = 0$, gives $z = 0$ or $\lambda = 1$, and $\lambda = -1$ from (1) forces $z = 0$. The lecture likewise says “$\lambda$ plus or minus 1” (T011a 00:13:10).

4. The theory lecture misreads the derivative at T011a 00:07:12 ("$yz$ plus $\lambda$ into $2z$"); it is $F_x = yz + 2\lambda x$. It also says "$x$ is equal to plus or minus 1" (T011a 00:08:28) for $x = \pm y$, and treats the volume as $8xyz$ and then $xyz$ (both give $x = y = z$).
5. Slide 011·4 lists, as a disadvantage, that the method cannot decide the nature of a stationary point. Added for clarity: in examination problems the nature follows from the problem (a largest box, a nearest point), or by comparing the values of $f$ at the stationary points (W2.4 compares 1 and 2).
6. CW problem 33's condition is garbled in the text layer; the rendered page reads $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$. Its answer $(a+b+c)^3$ at $(\frac{t}{a}, \frac{t}{b}, \frac{t}{c})$, $t = a+b+c$, is right (sympy, and a numerical check with $a, b, c = 1, 2, 3$).
7. No ESA answer key was used; every answer here is recomputed with sympy (May 2024 and Jul 2023 answer PDFs exist in the bucket but were not needed).
8. The theory lecture describes "a point in space" as "$x^2 + y^2 + z^2$ ... a sphere can be taken as a point" (T011a 00:09:35). Precisely: the squared distance from the origin is $x^2 + y^2 + z^2$, and minimising it is the same as minimising the distance (added for clarity).
9. Transcript recognition errors were corrected silently: "Lagrangius" / "Lagrangian's" / "Slagrange's" / "Lagrangus" / "grandiose" / "grandest" → Lagrange's (auxiliary function), "axillary" → auxiliary, "equity" / "equipment" → equations, "a Q" / "a q" → a cube, "two eight x" → $8xyz$, "Q" → cube, "constant" (for constraint) → constraint, "small problems" → solve problems.

## Coverage

| Source | Citations | Slides cited (slide × citations) |
|---|---|---|
| S011 | 44 | 3×9, 4×3, 5×4, 6×2, 7×6, 8×3, 9×4, 10×3, 11×2, 12×4, 13×2, 14×2 |
| CW | 5 | |
| ESA | 12 | |
| Extra | 5 | |
| T011a | 27 | |
| T011b | 11 | |

Slides 011·1, 011·2 and 011·15 are title, session and closing slides. Lecture T011a (theory, 14.8 min) works slides 3–9; lecture T011b (problems, 10.2 min) works slides 10–14. The deck title says "for two variables", but every problem uses three variables and one constraint.